

Pleural Diseases

The Big Picture

The pleura is a two-layer serous sac (**visceral** over lung, **parietal** lining chest wall) with a thin fluid film for frictionless breathing. **Only the parietal pleura feels pain** (intercostal nerves). Core disorders: **pneumothorax**, **pleural effusion/empyema**, **malignant effusion**, **chylothorax**, and **mesothelioma**.

Anatomy & Physiology

- Visceral pleura (lung + fissures) vs parietal (costal, mediastinal, diaphragmatic).
- **Parietal = somatic pain** (intercostal nn.); visceral/diaphragmatic = phrenic, **painless**.
- Normal fluid ~**0.25 mL/kg**; governed by **Starling forces**, filtered from parietal capillaries, reabsorbed by parietal lymphatics. Functions: reduce friction, couple lung to chest wall.

Pneumothorax

- Air in pleural space → lung collapse, ↓ compliance, V/Q mismatch.
- **Types**: primary spontaneous (PSP), secondary spontaneous (SSP — COPD, TB, CF, fibrosis), traumatic, iatrogenic (ventilation barotrauma, CVC, biopsy).
- **PSP**: ruptured apical **blebs/bullae**; **tall thin young males**, smoking; 80% have emphysema-like CT changes.
- **Tension pneumothorax** — air can't escape → mediastinal shift, hemodynamic collapse → **immediate needle decompression (2nd ICS midclavicular) then tube thoracostomy** (clinical diagnosis, no X-ray first).
- **Presentation**: sudden ipsilateral pleuritic pain + dyspnoea, ↓ breath sounds, hyperresonance. **Deep sulcus sign** on supine film.
- **Diagnosis**: upright **CXR** (CT gold-standard but not needed for first PSP).
- **Management**: small/asymptomatic → **observe + O₂** (re-expands ~2.2%/day, 79% success); aspiration; small percutaneous catheter; **large-bore tube thoracostomy** for trauma/unstable/large leak.
- **Recurrence 30% after first** (most <2 yrs). Surgery (**VATS bullectomy >95%**) for: persistent air leak >5 days, haemopneumothorax, recurrent/bilateral, first contralateral, at-risk jobs (pilots, divers). **Pleurodesis** (talco/bleomycin/doxycycline) an option.

Pleural Effusion & Empyema

- Effusion = abnormal fluid; **transudate (low protein) vs exudate (high protein)**. Benign 2× commoner than malignant.
- **Empyema = pus in pleural cavity**; precursor = pneumonia + parapneumonic effusion.
- **Causes**: complicated pneumonia, bronchiectasis, TB, lung abscess, trauma, oesophageal surgery, subphrenic abscess.
- **3 stages**: (1) exudative (low LDH, normal glucose/pH); (2) **fibropurulent** (loculations, fibrin); (3) **organising** (fibrous "**peel**" encasing lung — hardest to treat).
- **Light's exudate criteria**: pleural/serum protein >0.5, pleural/serum LDH >0.6, pleural LDH >■ upper normal, or **pH <7.0**.
- **Management**: antibiotics + thoracentesis work only for stage I; established empyema needs **drainage** (large-bore 32–38 Fr tube + **fibrinolytics**). Surgery (20–30%): **VATS debridement** (unified space + lung re-expansion), **decortication** via thoracotomy for stage III, open window if needed.

Malignant Pleural Effusion

- Up to **22%** of effusions; **95% metastatic** (lung + breast = 75%), also lymphoma, gastric, ovarian.
- Symptomatic: dyspnoea, cough, chest pain; ↓ tactile fremitus, dull percussion.
- Imaging: blunted costophrenic sulcus (as little as 125 mL), **nodular pleural thickening** on CT.
- Diagnosis: **thoracentesis (almost all exudates)** + cytology; pleural biopsy if cytology negative.
- Treatment: drain + **prevent recurrence with chemical pleurodesis** (talc, bleomycin, tetracycline).

Chylothorax

- **Lymph (chyle) in pleural space** — iatrogenic, traumatic, or spontaneous (thoracic duct).
- Manage: drainage + reduce chyle flow + nutrition; **early thoracic duct ligation** if no response (VATS).

Pleural Tumours (Mesothelioma)

- **Malignant pleural mesothelioma (MPM)** — commonest primary pleural tumour, aggressive, grave prognosis.
- **Asbestos exposure >80%**, ~20-yr latency; **5:1 male**, 6th–7th decade.
- Presentation: dyspnoea, chest-wall pain, malignant effusion, **restrictive pleural rind** (late).
- Work-up: **CT + FDG-PET**, MRI for chest wall depth; **thoracoscopic biopsy = gold standard (80%)**.
- Treatment: mostly advanced/unresectable; **extrapleural pneumonectomy** (en-bloc) in selected localised disease + multimodality chemo/radiotherapy.

Walk-Away Points

- **Only parietal pleura hurts**; PSP = tall thin young male smoker (apical blebs).
- **Tension pneumothorax = clinical → decompress immediately.**
- **Empyema = pus**; Light's criteria define exudate; stage III needs **decortication**.
- Malignant effusion = exudate, mostly metastatic (lung/breast) → **pleurodesis**.
- **Mesothelioma = asbestos**, 20-yr latency, thoracoscopic biopsy diagnostic.

Pleural Diseases — Revision Table	
Anatomy	Fact
Visceral pleura	Lung + fissures, painless (phrenic)
Parietal pleura	Chest wall, somatic pain (intercostal)
Normal fluid	~0.25 mL/kg, Starling forces
Pneumothorax	Detail
PSP	Apical blebs , tall thin young male smoker
SSP	COPD, TB, CF, fibrosis
Tension	Decompress 2nd ICS MCL → tube (clinical dx)
Sign	Deep sulcus (supine), hyperresonance
Diagnosis	Upright CXR
Small	Observe + O■ (2.2%/day, 79%)
Recurrence	30% after first (<2 yrs)
Surgery	VATS bullectomy >95% ; leak>5d, recurrent
Empyema	Detail
Definition	Pus in pleural cavity
Stages	Exudative → fibropurulent (loculations) → organising (peel)
Light's exudate	Protein>0.5, LDH>0.6, pH<7.0
Treatment	Tube (32–38Fr) + fibrinolytics; VATS / decortication
Malignant effusion	Detail
Frequency	22%; 95% metastatic (lung+breast 75%)
Type	Exudate
CT	Nodular pleural thickening
Treatment	Drain + pleurodesis (talc/bleomycin)
Chylothorax	Chyle in pleura; drain + thoracic duct ligation
Mesothelioma (MPM)	Detail
Cause	Asbestos >80% , 20-yr latency, 5:1 male
Diagnosis	Thoracoscopic biopsy (gold standard)
Surgery	Extrapleural pneumonectomy (selected)

Bold = high-yield.